S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

SentinelRisk AI

Explainable financial-event intelligence and portfolio risk simulation

Candidate: Bommaka Rajasekhar Reddy

IIT Dharwad

A prototype that turns unstructured text into traceable risk signals and illustrative portfolio actions.

The problem & our approach

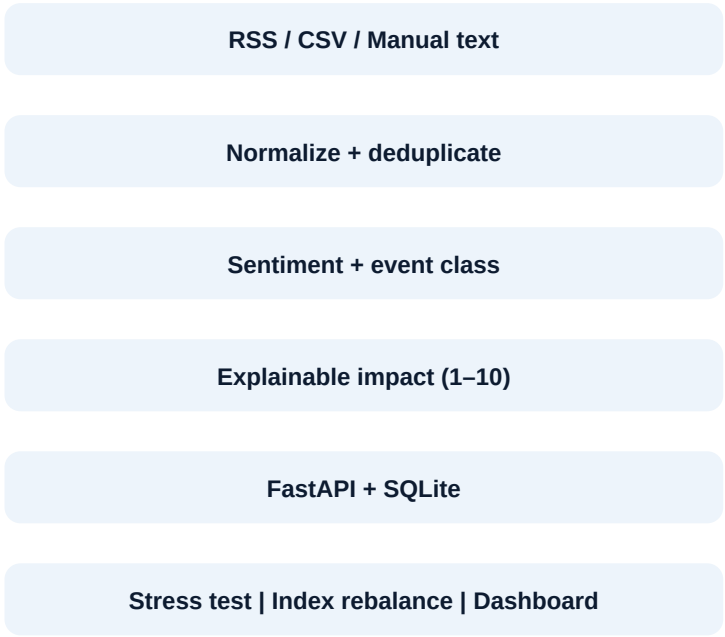
From fast-moving text to structured, decision-support signals

- Financial news and social-style text arrive in different formats and can be difficult to compare quickly.
- A headline alone does not show which event category matters, how severe it may be, or which exposures could be affected.
- SentinelRisk normalizes and deduplicates text, scores sentiment, classifies the event, and produces an explainable impact score (1–10).
- The same signal feeds two downstream demonstrations: portfolio stress testing and index rebalancing.

System design

One signal pipeline, two downstream workflows

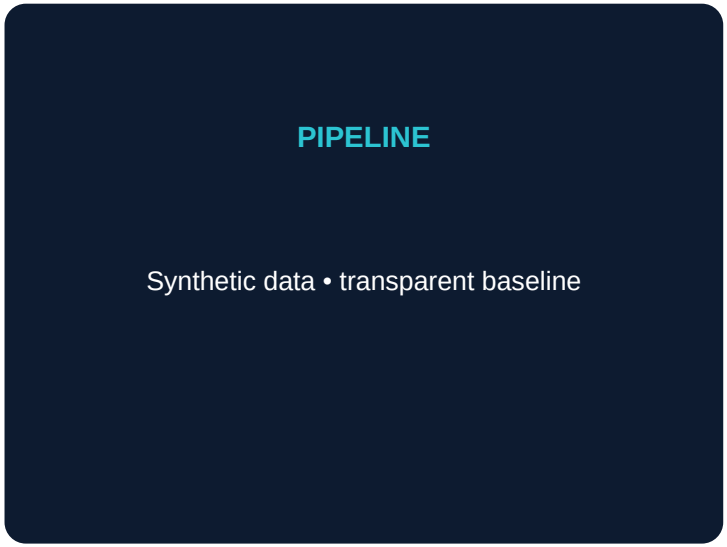
- Inputs: public RSS feeds (optional), bundled synthetic CSVs, and manual text.
- Pipeline: normalize → deduplicate → sentiment → event classification → explainable impact score.
- Serving and storage: FastAPI, SQLite, and a Streamlit dashboard.
- Outputs: risk signals, portfolio stress-test results, bounded index weights, and CSV/JSON exports.



Implementation highlights

Built to be reproducible, inspectable, and easy to demonstrate

- Python, pandas, NumPy, Pydantic, FastAPI, Streamlit, Plotly, and SQLite.
- Hybrid NLP design: transparent offline sentiment baseline by default; optional FinBERT sentiment mode.
- Event taxonomy includes geopolitical, macroeconomic, credit, M&A, regulatory/legal, earnings/business, and other.
- Signals preserve source/provenance fields; API endpoints support analysis, retrieval, and stress testing.
- The offline demo runs without API keys or a pretrained-model download.



Key results

Reproducible prototype checks on bundled synthetic data

- 18 unique risk signals generated in the demo run.
- Highest-impact event in that run: geopolitical, impact score 7/10.
- Illustrative synthetic portfolio value: 100.00 → 91.09 under the configured scenario.
- Rebalanced index weights sum to 1.000000.
- 11 automated tests passed in the previously verified project build; evaluation labels are small and synthetic.

18
unique signals

7/10
top event impact

100 → 91.09
scenario portfolio value

1.000000
index weight sum

Domain impact & limitations

A decision-support prototype—not an investment recommendation

- Potential value: help analysts triage event streams and trace an event to hypothetical exposure impacts.
- Stress testing makes assumptions visible; rebalancing enforces configured weight bounds and a 100% total.
- Data: all bundled headlines, posts, portfolio holdings, index constituents, and labelled evaluation examples are synthetic.
- Limitations: keyword-based baseline can miss context/negation; scenario shocks are illustrative, not calibrated forecasts.
- Not modelled fully: liquidity, default/recovery, convexity, options Greeks, or correlated market dynamics.

RESULTS ARE ILLUSTRATIVE

Synthetic data • transparent baseline

Demo, next steps & close

A clear path from prototype to stronger evaluation

- Live demo: run the offline pipeline, inspect a signal, execute a stress scenario, and view rebalanced weights.
- Next: evaluate on a larger independently labelled public dataset; calibrate impact scores and scenario assumptions.
- Next: improve entity linking, contextual NLP, monitoring, and deployment security.
- Repository: public GitHub, README quickstart, /src, /data, /docs/presentation.pdf, and architecture diagram.
- Thank you — questions?

THANK YOU

Questions & discussion